

Decision Trees Consolidation Activity

Pearson Edexcel IAL Business, Unit 3: Business decisions and strategy. Specification 3.3.3 (3) Decision trees. Case study: noon Minutes, UAE.

Name: _____	Class: _____	Date: _____
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Specification	Content
3 Decision trees	a) Construct and interpret simple decision-tree diagrams. b) Calculations and interpretations of figures generated by these techniques. c) Limitations of using decision trees.

Instructions: Read Extracts A to E before Section A. Read Extracts F and G before Section B. Show all workings. A decision node is a square, a chance node is a circle, and net gain = total expected value of an option minus its cost.

The case study: noon

Extract A: noon and noon Minutes

noon is an Emirati-founded digital platform, launched by Mohamed Alabbar in December 2017 in the United Arab Emirates and Saudi Arabia. It describes itself as the largest online shopping destination in the Middle East. Its quick commerce service, noon Minutes, delivers everything from groceries to gadgets in under 15 minutes in the UAE and Saudi Arabia, using small fulfilment centres called dark stores located near areas of high demand. noon One, its subscription, bundles free delivery and other benefits across noon's shopping, food and quick delivery services.

Extract B: The UAE quick commerce market

Quick commerce in the UAE has broadened from restaurant delivery to full-basket everyday retail. Leading players include Talabat, Instashop, noon, Careem and Carrefour. Industry analysts say noon's speed is a key differentiator, supported by a dense network of dark stores whose locations are guided by detailed local demand data. noon is considering how best to extend its 15-minute delivery promise to more customers across the UAE.

Extract C: Option A, new noon-owned dark stores

Under Option A, noon would open further dark stores of its own in areas outside its current coverage. noon would lease the sites, fit them out, recruit and train pickers and riders, and control the stock range and layout. This is a large up-front investment, and the stores would take time to find, fit out and staff. (Option A is a hypothetical option created for this exercise.)

Extract D: Option B, noon Minutes hubs in ADNOC service stations

ADNOC Distribution, the UAE's largest fuel and convenience retailer, has signed a Memorandum of Understanding (MoU) with noon to explore ways to improve last-mile delivery. Under it, new noon Minutes distribution centres would be placed within ADNOC Distribution service stations. ADNOC Distribution has 551 service stations and 373 ADNOC Oasis convenience stores across all seven emirates (as at 31 December 2024). ADNOC Oasis products ordered through the ADNOC Distribution app would be delivered by noon riders. An MoU is an agreement to explore co-operation, so the final terms are not yet fixed.

Extract E: Forecast data (illustrative)

noon has not published forecasts for these options. The figures below are invented for this exercise. Financial results are the estimated net cash inflows over five years, after running costs, in AED millions. A bracketed figure is a net cash outflow. Assume noon can pursue only one option, or do nothing (net gain AED 0).

	Option A: new noon-owned dark stores	Option B: noon Minutes hubs in ADNOC stations
Cost of the option (AED m)	85	50
Probability of strong demand	0.6	0.7
Financial result if demand is strong (AED m)	190	125
Probability of weak demand	0.4	0.3
Financial result if demand is weak (AED m)	(20)	30

Sources: ADNOC Distribution and noon announcement (via Zawya); Redseer Strategy Consulting; Research and Markets UAE quick commerce report (2026). Adapted for teaching; options, costs, probabilities and results are illustrative.

Section A: Construct and calculate

Task 1. Using the data in Extract E, construct a decision tree to show the options available to noon. Label the decision node, chance nodes, probabilities, costs and financial results. **(4 marks)**

Use a ruler and the space below.

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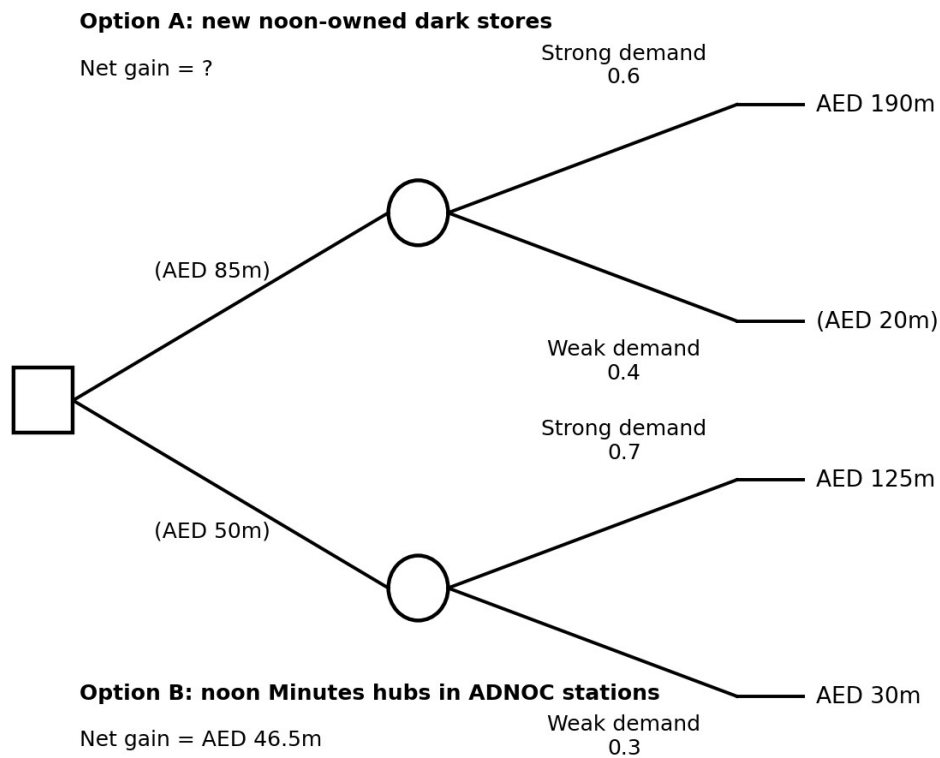
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(Total for Task 1 = 4 marks)

Task 2: Calculate the net gain

noon has drawn the decision tree below. The net gain for Option B is AED 46.5 million.



Using the data in the decision tree, calculate the net gain for Option A. You are advised to show your working. **(4 marks)**

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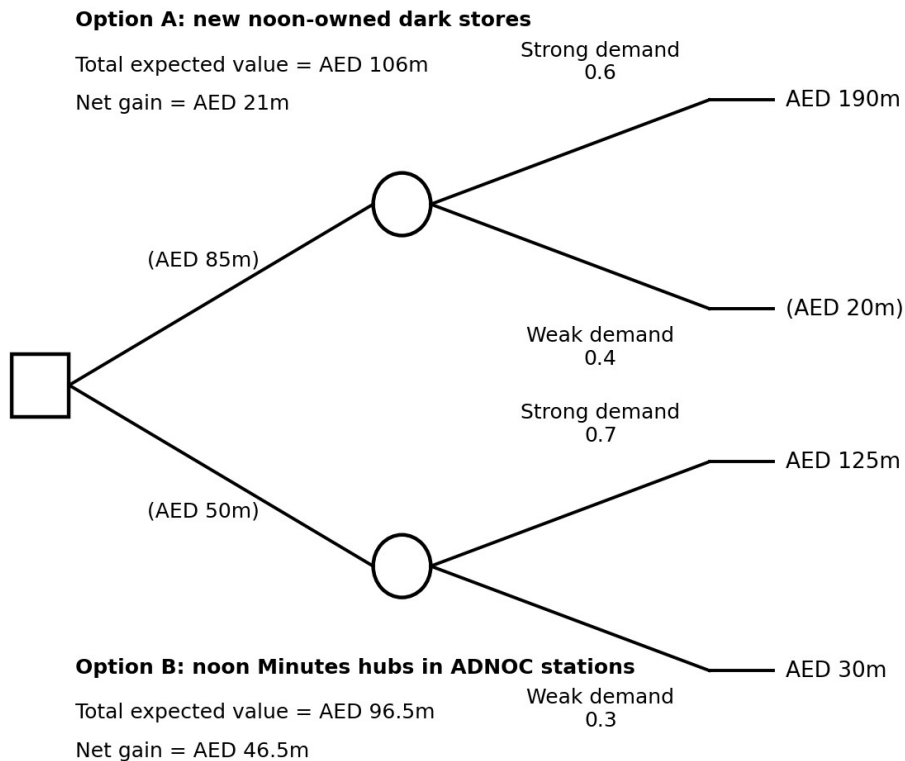
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(Total for Task 2 = 4 marks)

Section B: Interpret and evaluate

Read Extracts F and G, together with Extracts A to E, before answering Tasks 3 to 5.

Extract F: noon's completed decision tree (AED millions)



If noon does nothing, the net gain is AED 0.

Extract G: Other information

- Customers judge noon Minutes mainly on speed, and noon One subscribers expect free, fast delivery. Late or missing deliveries could damage loyalty.
- Under Option A, noon would control store layout, stock range and staff training, and would own the sites' fit-out and the customer data they generate.
- Under Option B, hubs inside service stations could open sooner and reach more customers, and would create new revenue for both noon and ADNOC Distribution. However, the agreement so far is an MoU to explore co-operation, so the terms, including space and sharing of income, are not final.
- Rivals such as Talabat, Careem and Carrefour could respond with their own offers, which could change demand for noon Minutes.
- Both options rely on estimates of customer demand. The probabilities in Extract E are managers' estimates, not guarantees.

Task 3. Assess the limitations of using decision trees to help noon decide between Option A and Option B. (8 marks)

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(Total for Task 3 = 8 marks)

Task 4: Usefulness of decision trees (12 marks)

Using the data in Extracts A to G, assess the usefulness of decision trees to noon when deciding how to expand noon Minutes. **(12 marks)**

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(Total for Task 4 = 12 marks)

Task 5: Evaluation (20 marks)

Evaluate which option noon should choose to extend its 15-minute delivery promise: Option A, new noon-owned dark stores, or Option B, noon Minutes hubs in ADNOC service stations. Use the decision tree and other information in your answer. **(20 marks)**

Plan first (5 minutes). Complete the grid before you write.

Evidence	Analysis (compare A and B directly)	However	Mini judgement
Net gain and total expected value			
Risk: best and worst outcomes			
Reliability of the probabilities			
Other factors (control, partner, competitors, speed)			

Level 4 checklist

- Use the figures: quote and compare net gains, and interpret them for noon.
- Develop the consequence for noon, not just the number.
- Challenge each argument with “However”, including how reliable the probabilities are.
- Add a mini judgement in every paragraph: which evidence carries more weight, and why.
- Finish with a conclusion that answers: Which option? Why? Why reject the other? What does success depend on? What extra evidence is needed?

Recommended writing time: 30 to 35 minutes.

(Total for Task 5 = 20 marks)

Stretch it: sensitivity analysis.

- The probability of strong demand for Option B falls from 0.7 to 0.3. Recalculate its net gain. Does your recommendation change?
- The result if demand is strong for Option A rises from AED 190m to AED 230m. Recalculate its net gain.
- The cost of Option A rises by 20%. Recalculate its net gain.
- What probability of strong demand for Option B would make its net gain the same as Option A's?